GIRIDHAR J SHENOY

Robotics Engineer — Manipulation, Motion Planning & Control | ROS 2 · C++ · Python

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https://giridhar-shenoy.github.io/portfolio-site/

PROFILE

Robotics engineer with 4+ years of experience across humanoid manipulation, motion planning, kinematics, dynamics, and control of industrial manipulators. Led manipulation planning and ROS2 system architecture for a humanoid robot at MyActuator in China, and developed gravity compensation and TCP jogging for 6-axis industrial robots at Orangewood Labs. Founded Function Robotics and built Bot & Brew, an autonomous robotic beverage kiosk, from concept to working hardware. Strong hands-on background in real robot systems, trajectory generation, depth-camera perception, hardware/software integration, Python/C++, and ROS2, with a focus on extending this classical robotics foundation into Physical AI and learning-based manipulation.

PUBLIC CODE (2026)

- github.com/giridhar-shenoy/ur5e-teaching-controllers — ros2_control effort controllers (hand guiding, TCP jogging, tracking) on a UR5e, C++17, two dynamics backends checked against each other; the hand-guiding law also run on a real AgileX Piper-H at zero driver stiffness (host-side torque at 100 Hz over CAN, gains identified on the arm)
- github.com/giridhar-shenoy/pointcloud-constrained-planning — MoveIt 2 constrained planning on a live octomap, plus a direct OMPL pipeline benchmarked against it on 100 queries, C++17
- github.com/giridhar-shenoy/sim-demo-to-policy — scripted demos → ACT and a spatial-softmax BC policy → 100-trial held-out evaluation

EMPLOYMENT HISTORY

Entrepreneur in Residence | Entrepreneur First, Bangalore
September 2026 – November 2026

- Explored robotics venture ideas in the Fall 2026 cohort; left to return to engineering.

AI Project Manager | Amrita Vishwa Vidyapeetham (Maitri Labs, AIMS Hospital), Kochi, India
March 2026 – August 2026

- Led day-to-day delivery for a medical AI research lab inside a 1,300+ bed quaternary care hospital, managing five ML engineers across oncology, hepatology, gastroenterology and neuro-rehabilitation projects with clinical collaborators.

Founder & Robotics Engineer | Function Robotics (Bot & Brew), Kochi, India
August 2025 – February 2026

- Founded Function Robotics and designed and built Bot & Brew, a fully automated robotic coffee kiosk, from scratch, owning mechanical integration, actuation, motion sequencing, control, and autonomous operation.
- Took the robot from concept to a working hardware prototype and demonstrated the full autonomous latte cycle end to end (video: https://drive.google.com/file/d/1dx2QZJ8dTxYjlFIyCTR4aFR4TP-qykYk/view).
- Built a zero-capex licensing model for F&B brands; incubated with Kerala Startup Mission and Maker Village.

Lead Robotics Software Engineer | MyActuator, China
April 2024 – July 2025

- Led manipulation-focused motion planning and control development for a humanoid robot, using OMPL-based constrained planning with depth-camera point clouds for motion around obstacles seen by the depth camera. Team video: https://www.youtube.com/watch?v=0hYw6lilqxE

- **Whole-arm manipulation:** built Cartesian path and trajectory generation with KDL, trapezoidal and spline velocity profiles, IK solvers, and Jacobian-based methods to convert task-space paths into joint-space motion.
- **ROS2 system architecture:** designed the high-level architecture integrating manipulation, navigation, and vision modules for the humanoid platform.
- **Vision-based motion retargeting:** built camera-based 3D skeletal tracking and geometric transformations that mapped human upper-body motion to robot joint angles in real time.

Robotics Control Engineer | Orangewood Labs, Delhi, India

June 2022 – March 2024

- Designed and shipped the gravity-compensated hand-guiding and Jacobian TCP-jogging controllers running on 30 production 6-axis arms.
- Built the joint-space PID trajectory-tracking controller used for programmed motion.
- **PLC interface:** built the interface between the robot control system and Siemens HMI and PLC over OPC-UA, enabling user interaction with robot I/Os and controllers.
- Collaborated across engineering teams, ran experimental validation of system performance, and conducted a vendor visit to China for actuator selection.

INTERNSHIPS

Robotics Intern | Orangewood Labs, Delhi, India

June 2021 – June 2022

- Developed sensor fusion, localization, SLAM, mapping, and navigation algorithms that enabled dynamic path planning for an autonomous mobile robot (AMR).

Hardware Intern | Riod Logic, Ernakulam, India

June 2020 – October 2020

- Designed and tested low-noise analog front ends for IoT sensing devices used in water-level detection and hydroponics monitoring, with LTspice simulation and cloud-connected deployments.

EDUCATION AND CERTIFICATION

B.Tech, Electrical and Electronics Engineering | Amrita Vishwa Vidyapeetham, Kollam, India

June 2018 – June 2022

Nanodegree in Robotics Software Engineering, Udacity

SKILLS

Robotics: Humanoid and industrial robot manipulation, motion planning (OMPL), trajectory generation, kinematics and dynamics (KDL), inverse kinematics and Jacobian methods, controller design, vision-based manipulation, robot hardware integration, path planning and navigation, SLAM, testing and validation

Software and Tools: ROS 2 (ros2_control, MoveIt 2), Gazebo, MuJoCo, Pinocchio, KDL, OMPL, LeRobot, PyTorch, Docker, GitHub Actions, MATLAB, LTspice

Languages and Protocols: Python, C++, CAN, EtherCAT, OPC-UA

Leadership: Technical team leadership, system architecture, cross-functional delivery, stakeholder communication, research-to-product translation

Focus: Learning-based manipulation: imitation learning from demonstration, data pipelines for robot learning (see sim-demo-to-policy)

Spoken Languages: English (fluent), Konkani (native), Hindi (fluent), Malayalam (fluent), Tamil (basic), Mandarin (learning)